

STAT 429: Introduction to Time Series Analysis

Homework 3

Fall 2026, University of Illinois Urbana-Champaign

Instructor: Sabyasachi Chatterjee

Due: TBA

Total Points = 104; Maximum Score = 100

1. Download the Google Trends Data (for the United States) for the query *mask*. This should be a monthly time series dataset that indicates the search popularity of this query from January 2004 to September 2025. The goal of this problem is to fit a smooth trend function $\{\mu_t\}$ to the data using the model:

$$y_t = \mu_t + \epsilon_t \quad \text{where } \epsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2)$$

and

$$\mu_t = \beta_0 + \beta_1(t-1) + \beta_2(t-2)_+ + \beta_3(t-3)_+ + \cdots + \beta_{n-1}(t-(n-1))_+.$$

- a) Fit the trend function μ_t to the data using ridge regularization with tuning parameter $\lambda = 1, 10, 100$, and 1000. Compare and interpret the differences among these four trend estimates. Which of these estimates provides the best summary for the data? **(4 points)**
 - b) Fit the trend function μ_t to the data using LASSO regularization with tuning parameter $\lambda = 1, 10, 100$, and 1000. Compare and interpret the differences among these four trend estimates. Which of these estimates provides the best summary for the data? **(4 points)**
 - c) Among the eight estimated trend functions (four from ridge and four from LASSO), which one is the most suitable for this dataset? Explain your reasoning. **(2 points)**
2. Download the FRED dataset on the federal minimum wage hourly wage for non-farm workers for the United States from <https://fred.stlouisfed.org/series/FEDMINNFRWG>. This is a monthly dataset (from Oct 1938 to Aug 2025). The goal of this problem is to fit a trend function $\{\mu_t\}$ to the data using the model:

$$y_t = \mu_t + \epsilon_t \quad \text{where } \epsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2)$$

and

$$\mu_t = \beta_0 + \beta_1 I\{t > 1\} + \beta_2 I\{t > 2\} + \beta_3 I\{t > 3\} + \cdots + \beta_{n-1} I\{t > n-1\}.$$

Here $I\{t > j\}$ equals 1 if $t > j$ and 0 otherwise. Ridge regularization here would use the penalty $\lambda \sum_{j=1}^{n-1} \beta_j^2$ and LASSO regularization would use the penalty $\lambda \sum_{j=1}^{n-1} |\beta_j|$.

- a) Fit the trend function μ_t to the data using ridge regularization with tuning parameter $\lambda = 1, 10, 100$, and 1000 . Compare and interpret the differences among these four trend estimates. Which of these estimates provides the best summary for the data? **(4 points)**
- b) Fit the trend function μ_t to the data using LASSO regularization with tuning parameter $\lambda = 1, 10, 100$, and 1000 . Compare and interpret the differences among these four trend estimates. Which of these estimates provides the best summary for the data? **(4 points)**
- c) Among the eight estimated trend functions (four from ridge and four from LASSO), which one is the most suitable for this dataset? Explain your reasoning. **(2 points)**
3. Consider a univariate regression dataset $(x_1, y_1), \dots, (x_n, y_n)$ where each x_i and y_i are real-valued. Fix $\lambda > 0$.

- a) Suppose $(\hat{\beta}_0^{\text{ridge}}(\lambda), \hat{\beta}_1^{\text{ridge}}(\lambda))$ minimize

$$\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 + \lambda \beta_1^2$$

over all β_0, β_1 . Show that **(3 points)**

$$\hat{\beta}_1^{\text{ridge}}(\lambda) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\lambda + \sum_{i=1}^n (x_i - \bar{x})^2} \quad \text{and} \quad \hat{\beta}_0^{\text{ridge}}(\lambda) = \bar{y} - \bar{x} \hat{\beta}_1^{\text{ridge}}(\lambda).$$

- b) Suppose $(\hat{\beta}_0^{\text{lasso}}(\lambda), \hat{\beta}_1^{\text{lasso}}(\lambda))$ minimize

$$\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 + \lambda |\beta_1|$$

over all β_0, β_1 . Show that $\hat{\beta}_0^{\text{lasso}}(\lambda) = \bar{y} - \bar{x} \hat{\beta}_1^{\text{lasso}}(\lambda)$ and **(5 Points)**

$$\hat{\beta}_1^{\text{lasso}}(\lambda) = \begin{cases} \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) - \lambda/2}{\sum_{i=1}^n (x_i - \bar{x})^2} & \text{if } \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) > \lambda/2 \\ \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) + \lambda/2}{\sum_{i=1}^n (x_i - \bar{x})^2} & \text{if } \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) < -\lambda/2 \\ 0 & \text{if } -\lambda/2 \leq \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) \leq \lambda/2. \end{cases}$$

4. Download the Google Trends Data (for the United States) for the query *yahoo*. This should be a monthly time series dataset that indicates the search popularity of this query from January 2004 to September 2025. The goal of this problem is to fit a smooth trend function $\{\mu_t\}$ to the data using the model:

$$y_t = \mu_t + \epsilon_t \quad \text{where } \epsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2)$$

and

$$\mu_t = \beta_0 + \beta_1(t-1) + \beta_2(t-2)_+ + \beta_3(t-3)_+ + \dots + \beta_{n-1}(t-(n-1))_+.$$

- a) Implement ridge regression (with a cross validation strategy for determining the regularization parameter) to estimate μ_t . Report your value of λ and plot the estimated trend function $\hat{\mu}_t$ on a plot of the data. Comment on whether the fitted trend function is capturing well the patterns present in the data without fully interpolating the data. **(6 points)**

b) Consider Bayesian inference with the prior:

$$\beta_0, \beta_1 \stackrel{\text{i.i.d}}{\sim} N(0, C), \quad \beta_2, \dots, \beta_{n-1} \stackrel{\text{i.i.d}}{\sim} N(0, \tau^2)$$

along with $\text{unif}(-C, C)$ priors for τ and σ (C is a large number).

- i. Calculate point estimates $\hat{\tau}$ of τ and $\hat{\sigma}$ of σ . How does your choice of the tuning parameter λ in part (a) compare to $\hat{\sigma}^2/\hat{\tau}^2$? (**4 points**).
 - ii. Draw $N = 1000$ posterior samples from $\{\mu_t\}$ and plot these sampled trend functions on a plot of the data. Comment on whether the uncertainty revealed in this plot makes sense. (**4 points**)
 - iii. Average over the N posterior samples to obtain the Bayesian point estimate of $\hat{\mu}_t$. Compare this with the ridge regression estimate from part (a). Which trend estimate is smoother? Which trend estimate would you prefer? (**4 points**).
5. Download the Google Trends Data (for the United States) for the query *golf*. This should be a monthly time series dataset that indicates the search popularity of this query from January 2004 to September 2025. The goal of this problem is to fit a smooth trend function $\{\mu_t\}$ to the data using the model:

$$y_t = \mu_t + \epsilon_t \quad \text{where } \epsilon_t \stackrel{\text{i.i.d}}{\sim} N(0, \sigma^2)$$

and

$$\mu_t = \beta_0 + \beta_1(t-1) + \beta_2(t-2)_+ + \beta_3(t-3)_+ + \dots + \beta_{n-1}(t-(n-1))_+.$$

- a) Implement ridge regression (with a cross validation strategy for determining the regularization parameter) to estimate μ_t . Report your value of λ and plot the estimated trend function $\hat{\mu}_t$ on a plot of the data. Comment on whether the fitted trend function is capturing well the patterns present in the data without fully interpolating the data. (**6 points**)
- b) Consider Bayesian inference with the prior:

$$\beta_0, \beta_1 \stackrel{\text{i.i.d}}{\sim} N(0, C), \quad \beta_2, \dots, \beta_{n-1} \stackrel{\text{i.i.d}}{\sim} N(0, \tau^2)$$

along with $\text{unif}(-C, C)$ priors for τ and σ (C is a large number).

- i. Calculate point estimates $\hat{\tau}$ of τ and $\hat{\sigma}$ of σ . How does your choice of the tuning parameter λ in part (a) compare to $\hat{\sigma}^2/\hat{\tau}^2$? (**4 points**).
 - ii. Draw $N = 1000$ posterior samples from $\{\mu_t\}$ and plot these sampled trend functions on a plot of the data. Comment on whether the uncertainty revealed in this plot makes sense. (**4 points**)
 - iii. Average over the N posterior samples to obtain the Bayesian point estimate of $\hat{\mu}_t$. Compare this with the ridge regression estimate from part (a). Which trend estimate is smoother? Which trend estimate would you prefer? (**4 points**).
6. Consider again the *golf* google trends dataset as in the previous problem. In this problem, we will consider the model

$$y_t = \beta_0 + \beta_1(t-1) + \beta_2(t-2)_+ + \beta_3(t-3)_+ + \dots + \beta_{n-1}(t-(n-1))_+ \\ + \beta_n \cos\left(2\pi \frac{t}{12}\right) + \beta_{n+1} \sin\left(2\pi \frac{t}{12}\right) + \epsilon_t.$$

a) A natural ridge regression estimator here would minimize:

$$\sum_{t=1}^n \left(y_t - \beta_0 - \beta_1(t-1) - \beta_2(t-2)_+ - \beta_3(t-3)_+ - \cdots - \beta_{n-1}(t-(n-1))_+ - \beta_n \cos\left(2\pi \frac{t}{12}\right) - \beta_{n+1} \sin\left(2\pi \frac{t}{12}\right) \right)^2 + \lambda \sum_{j=2}^{n-1} \beta_j^2$$

Implement this estimator with a cross validation strategy for determining the regularization parameter. Report your value of λ and plot the estimated trend functions:

$$\hat{\mu}_t := \hat{\beta}_0 + \hat{\beta}_1(t-1) + \hat{\beta}_2(t-2)_+ + \hat{\beta}_3(t-3)_+ + \cdots + \hat{\beta}_{n-1}(t-(n-1))_+$$

and

$$\hat{\mu}_t + \hat{\beta}_n \cos\left(2\pi \frac{t}{12}\right) + \hat{\beta}_{n+1} \sin\left(2\pi \frac{t}{12}\right).$$

on a plot of the data. Comment on whether these trend functions capture well the patterns present in the data without fully interpolating the data. **(8 points)**

b) Consider Bayesian inference with the prior:

$$\beta_0, \beta_1, \beta_n, \beta_{n+1} \stackrel{\text{i.i.d}}{\sim} N(0, C), \quad \beta_2, \dots, \beta_{n-1} \stackrel{\text{i.i.d}}{\sim} N(0, \tau^2)$$

along with $\text{unif}(-C, C)$ priors for τ and σ (C is a large number).

- i. Calculate point estimates $\hat{\tau}$ of τ and $\hat{\sigma}$ of σ . How does your choice of the tuning parameter λ in part (a) compare to $\hat{\sigma}^2/\hat{\tau}^2$? **(5 points)**.
- ii. Draw $N = 1000$ posterior samples from $\{\mu_t\}$ (note that μ_t does not include the sinusoidal part) and plot these sampled trend functions on a plot of the data. Comment on whether the uncertainty revealed in this plot makes sense. **(5 points)**
- iii. Average over the N posterior samples to obtain the Bayesian point estimate of $\hat{\mu}_t$. Compare this with the ridge regression estimate from part (a). Which trend estimate is smoother? Which trend estimate would you prefer? **(5 points)**.